G12Adv	Mathematics	الرياضيات	T1 - 2018/19

Multiple Choice Part	جزء الاختيار من متعدد
Show all your work when answering these questions.	يجب كتابة خطوات الحل التفصيلية للمفردات الاختبارية كافة.

Multiple Choice — اختيار من متعدد

1. **Evaluate**

$$\lim_{x\to 0} \frac{x}{3-\sqrt{x+9}}$$

A) $\frac{1}{3}$ **B)** 0

C) -6 **D)** Does not exist

2. **Find** $\lim_{h\to 0} \frac{(2+h)^3-8}{h}$ **if it exists.**

A) 4 **B)** 8

C) 12 **D)** Does not exist

3. **If** $f(x)=x^4+3x^2-2$, **find** $f'''\left(\frac{1}{6}\right)$.

A) $f'''\left(\frac{1}{6}\right)=2$

B) $f'''\left(\frac{1}{6}\right)=-4$

C) $f'''\left(\frac{1}{6}\right)=4$

D) $f'''\left(\frac{1}{6}\right)=-2$

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4. **Find the derivative of the function** $f(x) = \cos^{-1}(2x)$.

A) $f'(x) = \dfrac{2\sin(2x)}{\cos^2(x-2)}$

B) $f'(x) = \dfrac{2}{\sqrt{1-4x^2}}$

C) $f'(x) = \dfrac{-2}{\sqrt{1+4x^2}}$

D) $f'(x) = \dfrac{-2}{\sqrt{1-4x^2}}$

5. **Find all values of** x **for which the tangent line to the curve** $y = x^3 - 6x^2 + 1$ **is horizontal.**

A) $x = 0,\ x = 4$

B) $x = -4,\ x = 0,\ x = 4$

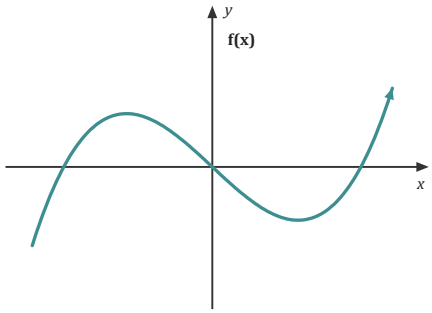
C) $x = 0,\ x = -4$

D) $x = -1,\ x = 0,\ x = 1$

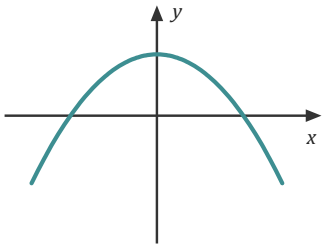
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Multiple Choice	اختيار من متعدد

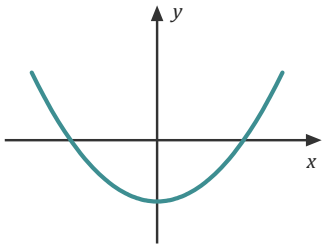
6. Use the graph of $y = f(x)$ to sketch a graph of $y = f''(x)$.



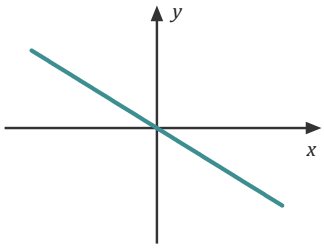
A)



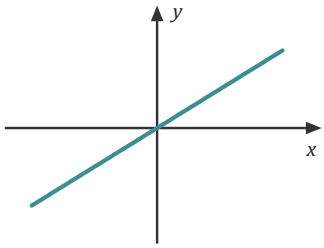
B)



C)



D)



7. Find $\lim_{x \to 2} \dfrac{x^3 - 8}{2x^2 - 3x - 2}$ if it exists.

A) $-\dfrac{5}{12}$

B) $-\dfrac{12}{5}$

C) $\dfrac{12}{5}$

D) $\dfrac{5}{12}$

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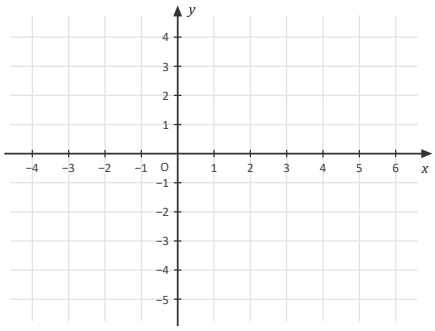
8. **Determine the values of a and b that make the given function continuous.**

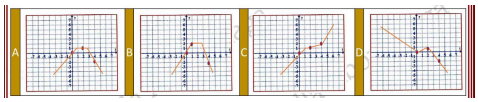
$$f(x)=\begin{cases} a(\tan^{-1}x+2), & x<0 \\ b\cos\left(\left(x+\frac{1}{3}\right)\pi\right), & 0\le x\le 3 \\ \ln(x-2)+x^2+1, & x>3 \end{cases}$$

A) $a=-5,\ b=-20$ **B)** $a=5,\ b=20$

C) $a=-5,\ b=20$ **D)** $a=5,\ b=-20$

9. **Sketch the graph of a function with the following properties:** $f(0)=0,\ f(2)=1,\ f(4)=-2,\ f'(0)=1,\ f'(2)=0$ **and** $f'(4)=-3.$





A)

B)

C)

D)

10. **Find** $\lim\limits_{x\to 2}\dfrac{x-2}{|x-2|}$ **if it exists.**

A) -2 **B)** -1

C) 0 **D)** Does not exist

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11. Find $y'(x)$ implicitly for $xy^2 + 5x = (2y+1)^3$.

A) $y'(x) = \dfrac{y^2 - 5}{2xy - 6(2y+1)^2}$

B) $y'(x) = \dfrac{-y^2 - 5}{2xy - 6(2y+1)^2}$

C) $y'(x) = \dfrac{-y^2 + 5}{2xy - 6(2y+1)^2}$

D) $y'(x) = \dfrac{-y^2 - 5}{2xy + 6(2y+1)^2}$

12. Find the derivative of the function $f(x) = \dfrac{e^{\sqrt{x^3+1}}}{2x}$.

A) $f'(x) = \dfrac{\left[2x\left(\dfrac{3x^2}{2\sqrt{x^3+1}}\right)e^{\sqrt{x^3+1}}\right] + \left[e^{\sqrt{x^3+1}}\right]}{(2x)^2}$

B) $f'(x) = \dfrac{\left[3x\left(\dfrac{2x^2}{2\sqrt{x^3+1}}\right)e^{\sqrt{x^3+1}}\right] - \left[2e^{\sqrt{x^3+1}}\right]}{(2x)^2}$

C) $f'(x) = \dfrac{\left[2x\left(\dfrac{3x^2}{\sqrt{x^3+1}}\right)e^{\sqrt{x^3+1}}\right] - \left[2e^{\sqrt{x^3+1}}\right]}{(2x)^2}$

D) $f'(x) = \dfrac{\left[2x\left(\dfrac{3x^2}{2\sqrt{x^3+1}}\right)e^{\sqrt{x^3+1}}\right] - \left[2e^{\sqrt{x^3+1}}\right]}{(2x)^2}$

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13. **Use logarithmic differentiation to find the derivative of** $f(x) = \left(\frac{1}{3}\right)^{x^2}$.

A) $h'(x) = \left(\frac{1}{3}\right)^{x^2}\left(x \ln \frac{1}{3}\right)$

B) $h'(x) = \left(\frac{1}{3}\right)^{x}\left(-2x \ln \frac{1}{3}\right)$

C) $h'(x) = \left(\frac{1}{3}\right)^{x^2}(2x \ln 3)$

D) $h'(x) = \left(\frac{1}{3}\right)^{x^2}\left(2x \ln \frac{1}{3}\right)$

14. **A spring hanging from the ceiling vibrates up and down. Its vertical position at time** $0 \leq t \leq \pi$ **is given by** $f(t) = 4\cos(2t)$**. What is its location when its velocity is 0?**

A) 4 and -4 **B)** 2 and -2

C) 8 and -8 **D)** 5 and -5

15. **(b) Show that the function** $f(x) = 3 - x + e^{-x}$ **is decreasing.**

A) Increasing because F'(x)=1-e^{-x}>0

B) Decreasing because F'(x)=1+e^{-x}<0

C) Increasing because F'(x)=-1-e^{-x}>0

D) Decreasing because F'(x)=-1-e^{-x}<0

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16. Determine the interval(s) where $f(x) = \dfrac{\ln(x^2-1)}{\sqrt{x^2-2x}}$ is continuous.

A) $(-\infty, -1] \cup [2, \infty)$

B) $(-\infty, 0) \cup (2, \infty)$

C) $(-\infty, -1) \cup (2, \infty)$

D) $(-\infty, -1) \cup (1, \infty)$

17. Find $\lim\limits_{x\to 2} \dfrac{x-2}{|x-2|}$ if it exists.

A) -2

B) -1

C) 0

D) Does not exist

18. Evaluate

$$\lim_{x\to\infty} \cot^{-1} x$$

A) 0

B) $-\dfrac{\pi}{2}$

C) 1

D) ∞

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19. **Determine the interval where the function $f(x) = x^2 - x + 1$ satisfies Rolle's theorem and find the value of c.**

A) $[-1, 1],\ c = \frac{1}{2}$

B) $[0, 1],\ c = \frac{1}{2}$

C) $[-2, 2],\ c = 0$

D) $[0, 1],\ c = 2$

20. **Evaluate**

$$\lim_{x \to 0} \left(\frac{x\, e^{-2x+1}}{x^2 + x} \right)$$

A) 0 **B)** 1

C) e **D)** e^2